Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Total heat load inside a control enclosure, the effective surface that can dissipate it for the mounting condition, the internal temperature rise with natural convection and radiation through a stated overall heat-transfer coefficient, the forced-ventilation airflow for a target rise (density corrected for altitude), the fan duty point against a fan curve and the system resistance, optional solar gain, and the cooling-unit capacity needed when the limit cannot be met.
$P = \Sigma P_i n_i d_i (+ \text{solar } \alpha I A_{sun})$. $A_{eff} = \Sigma f_{face} \times A_{face}$. Natural: $\Delta T = P/(h A_{eff})$ (a plain sum of exposed face areas times one coefficient; the IEC TR 60890 tables are not used). Forced: $V = P/(\rho c_p \Delta T_{target})$ with $\rho = p(z)/(287.055\ T)$ and $c_p = 1006$ J/kg·K; fan duty where $p_{fan}(V) = p_{sys} (V/V_{ref})^2$ (bisection). Cooling unit: $P_{cool} = P - h A_{eff} (T_{max} - T_{amb})$ when positive. Sealed enclosure (T_{seal} entered, no breather): the pressure difference across the walls when the air warms from the sealing temperature to T_{int} at constant volume, $\Delta p = p_{site} \cdot |T_{int}/T_{seal} - 1|$ (absolute temperatures), a rise when $T_{int} > T_{seal}$, a suction when below. A pressure-compensation element (breather or relief) limits it: $dp_{seal} = \min(\text{that difference}, dp_{relief})$, the cracking pressure. The force of dp_{seal} on a cover of area A_{cover} is $F_{seal} = dp_{seal} \cdot A_{cover}$.
Ref: IEC TR 60890 — A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation
Ref: Manufacturer enclosure climate-control guides (Rittal, nVent Hoffman) — Effective surface, heat-transfer coefficient and fan-sizing practice (cited, not copied)
Ref: BS EN 61439-1 10.10 — Verification of temperature rise

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Heat loads	2 row(s)		spec: Panel schedule PNL-09	any
loads[1]		name = PLC and I/O; P = 80 W; qty = 1; duty = 1			
loads[2]		name = drives; P = 55 W; qty = 4; duty = 1			
P_add	Additional heat load (linked from E6's dissipated heat or another calculation)	0	W	default	0 … 100000 W
I_sun	Solar irradiance on the enclosure (0 = indoors)	800	W/m^2	spec: Site conditions SC-5: 800 W/m² on the sunlit side	0 … 1200 W/m^2
A_sun	Sunlit area	0.48	m^2	assumption: One 600 × 800 face in the sun.	0 … 100 m^2

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
alpha_sun	Surface absorptivity	0.5		assumption: Light-grey paint.	0.1 … 1
W	Enclosure width	0.6	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
H	Enclosure height	0.8	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
D	Enclosure depth	0.3	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
f_top	Top exposure (1 free, 0 covered)	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_front	Front exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_rear	Rear exposure (0 = wall-mounted)	1		assumption: Free-standing.	0 … 1
f_left	Left side exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_right	Right side exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_bottom	Bottom exposure (0 = on the floor)	0		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
h	Overall heat-transfer coefficient of the enclosure surface (editable reference value)	5.5	W/(m^2*K)	reference: Reference record: painted steel enclosure, still air, 5.5 W/m²K	3 … 8 W/(m^2*K)
T_amb	Ambient temperature	40	degC	spec: Site conditions SC-5	-20 … 55 degC
T_max	Maximum internal temperature	45	degC	spec: Site conditions SC-5	20 … 80 degC
altitude	Site altitude (air density)	0	m	assumption: Free-standing kiosk, bottom on a plinth.	0 … 4000 m
p_sys	System resistance (filters, grilles) at the reference airflow	20	Pa	default	0 … 500 Pa
V_ref	Reference airflow for the system resistance	100	m^3/h	default	1 … 100000 m^3/h

3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h (natural convection plus radiation to surroundings at the ambient temperature); covered faces dissipate nothing. (IEC TR 60890-style effective-surface approach; h is a reference value the user may edit.)
A2. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise; natural dissipation is neglected for the airflow (conservative). (Manufacturer fan-sizing practice.)
A3. A sealed enclosure (T_seal entered) is a closed, rigid, leak-free volume of dry air at the site pressure when closed: the pressure difference follows from the ideal-gas relation at constant volume, from the sealing temperature to the internal temperature with natural dissipation T_int (the temperature without a cooling unit); a breather, a seal that leaks or a flexing wall reduces it, so it is an upper bound (a pressure-compensation element is modelled only as a cap at its cracking pressure, with no flow resistance); the site pressure is the standard atmosphere at the altitude. (Ideal gas at constant volume (derived).)
A4. The fan curve is linear between the entered points; the system resistance scales with airflow squared. (Duty-point idealisation.)
A5. A_sun (Sunlit area) is taken as 0.48 m^2. (One 600 × 800 face in the sun.)
A6. alpha_sun (Surface absorptivity) is taken as 0.5. (Light-grey paint.)
A7. f_top (Top exposure (1 free, 0 covered)) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A8. f_front (Front exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A9. f_rear (Rear exposure (0 = wall-mounted)) is taken as 1. (Free-standing.)
A10. f_left (Left side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A11. f_right (Right side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A12. f_bottom (Bottom exposure (0 = on the floor)) is taken as 0. (Free-standing kiosk, bottom on a plinth.)
A13. altitude (Site altitude (air density)) is taken as 0 m. (Free-standing kiosk, bottom on a plinth.)

4. Working

[18] P_loads = ∑ P_i × n_i × d_i
= 2 item(s)
= 300 W
Internal heat load

[19] P_solar = alpha_sun * I_sun * A_sun
= 0.5 × 800 W/m² × 0.48 m²
= 192 W
Solar gain

[20] P_total = P_loads + P_add + P_solar
= 300 W + 0 W + 192 W
= 492 W
Total heat to remove

[21] A_eff = f_top * W * D + f_bottom * W * D + f_front * W * H + f_rear * W * H + f_left * H * D
IEC TR 60890 approach
= 1 × 0.6 m × 0.3 m + 0 × 0.6 m × 0.3 m + 1 × 0.6 m × 0.8 m + 1 × 0.6 m × 0.8 m + 1 × 0.8 m × 0.8 m × 0.3 m
= 1.62 m²
Effective dissipating surface

[22] dT_nat = P_total / (h * A_eff)
= 492 W / (5.5 W/(m²·K) × 1.62 m²)
= 55.219 K
Temperature rise with natural dissipation

[23] T_int = T_amb + dT_nat
= 40 degC + 55.219 K
= 95.219 degC
Internal temperature with natural dissipation

[24] dT_allow = T_max - T_amb
= 45 degC - 40 degC
= 5 K
Allowed rise
[25] u_nat = dT_nat / dT_allow
= 55.219 K / 5 K
= 11.044
Natural dissipation utilisation
[26] P_nat_max = h * A_eff * dT_allow
= 5.5 W/(m²·K) × 1.62 m² × 5 K
= 44.55 W
Heat the enclosure dissipates naturally at the limit
[27] P_cool = max(0 W, P_total - P_nat_max)
= max(0 W, 492 W - 44.55 W)
= 447.45 W
Cooling-unit capacity needed if ventilation is not used
[28] p_site = 101325 (1 - 2.25577e-5 z)^5.2559
= z = 0 m
= 101.32 kPa
Site pressure
[29] rho_air = p_site / (287.055 J/(kg*K) * T_amb)
= 101.32 kPa / (287.055 J/(kg*K) × 40 degC)
= 1.1272 kg/m³
Air density at ambient and altitude
[30] V_req = P_total / (rho_air * 1006 J/(kg*K) * dT_allow)
= 492 W / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K)
= 312.39 m³/h
Airflow for forced ventilation at the allowed rise (natural dissipation neglected)
[31] V_req_net = max(0 m^3/h, (P_total - P_nat_max) / (rho_air * 1006 J/(kg*K) * dT_allow))
= max(0 m^3/h, (492 W - 44.55 W) / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K))
= 284.11 m³/h
Airflow crediting natural dissipation

5. Results

SYMBOL	VALUE	DESCRIPTION
P_loads	300 W	Internal heat load
P_solar	192 W	Solar gain
P_total	492 W	Total heat
A_eff	1.62 m²	Effective surface
dT_nat	55.219 K	Natural temperature rise
T_int	95.219 degC	Internal temperature (natural)
P_nat_max	44.55 W	Natural dissipation at the limit
P_cool	447.45 W	Cooling unit needed
T_max	45 degC	Internal temperature limit
rho_air	1.1272 kg/m³	Air density
V_req	312.39 m³/h	Required airflow
V_req_net	284.11 m³/h	Required airflow (net)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Natural dissipation holds T_max	u_nat = 11.044	u_nat ≤ 1	11	

7. Applicability

WARNING — input.P_add: P_add (Additional heat load (linked from E6's dissipated heat or another calculation)) was not entered; the template default 0 W is used.
WARNING — input.p_sys: p_sys (System resistance (filters, grilles) at the reference airflow) was not entered; the template default 20 Pa is used.
WARNING — input.V_ref: V_ref (Reference airflow for the system resistance) was not entered; the template default 100 m^3/h is used.
WARNING — cooling-unit: T_max: natural dissipation holds 44.55 W of 492 W; ventilate or fit a 447.4 W cooling unit.

8. Verification

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Template elec.enclosure-heat@1.4.0 Content hash 3add357ba3b44d85 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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